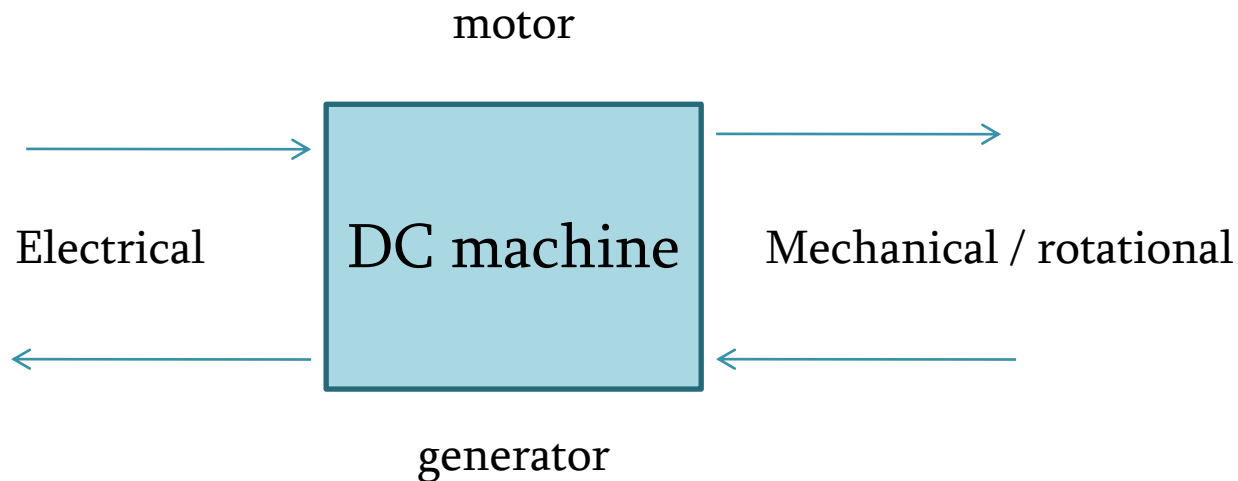


DC MACHINES FUNDAMENTALS

1. Describe, identify and explain the construction and operations, armature windings and commutation
2. Identify and explain induced voltage and developed torque.
3. Define the power flow and losses.
4. Explain the problems of commutation and their solutions

1. Construction and operations, armature windings and commutation

A DC Machines can be used as either a DC generator or a DC motor.



DC generators:

- To convert mechanical energy to electrical energy.
- Limited use due to solid state rectifier.

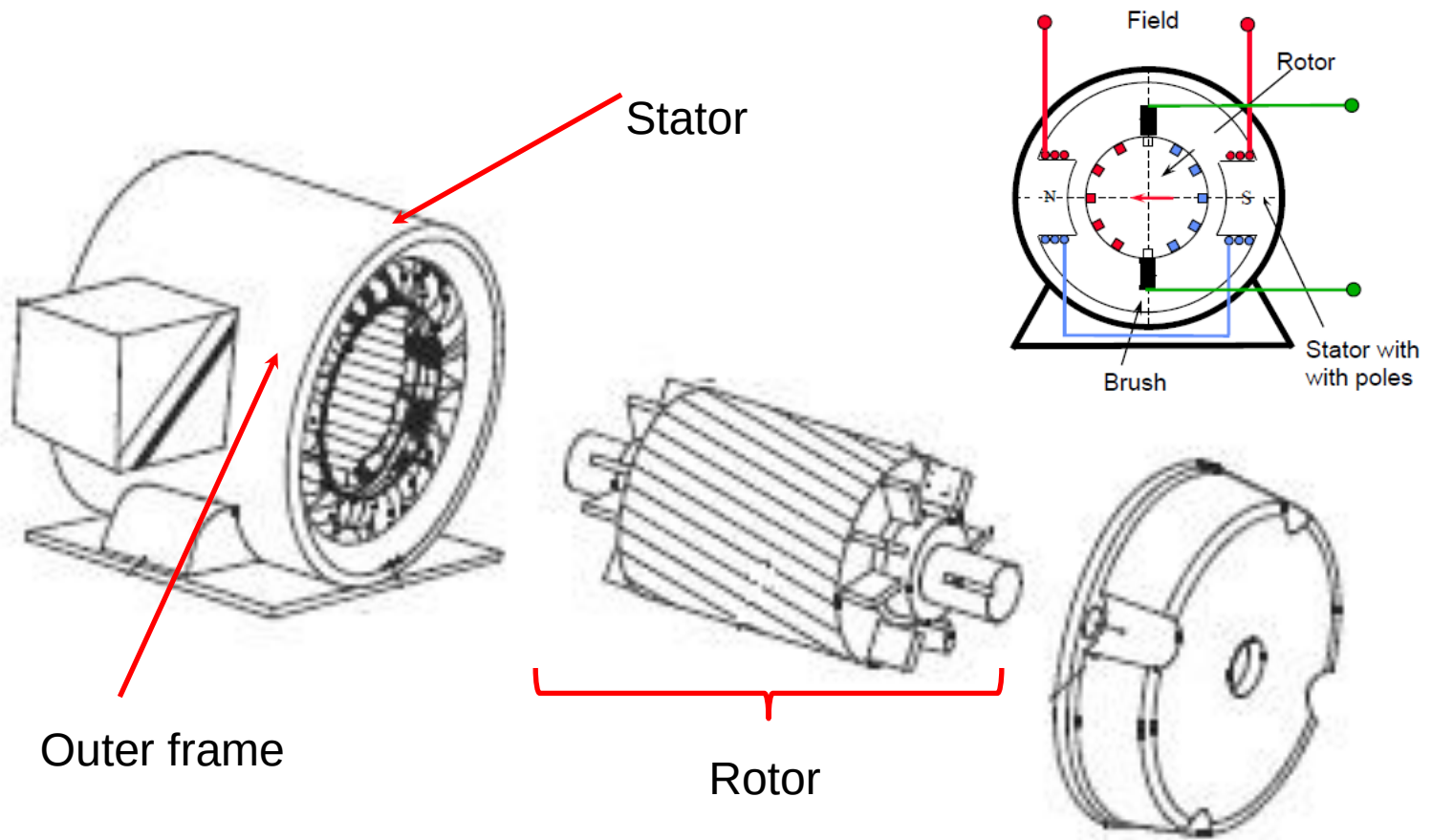
DC motors:

- To convert electrical energy to mechanical energy
- Widely used
- Main feature: speed control is simple and cheap

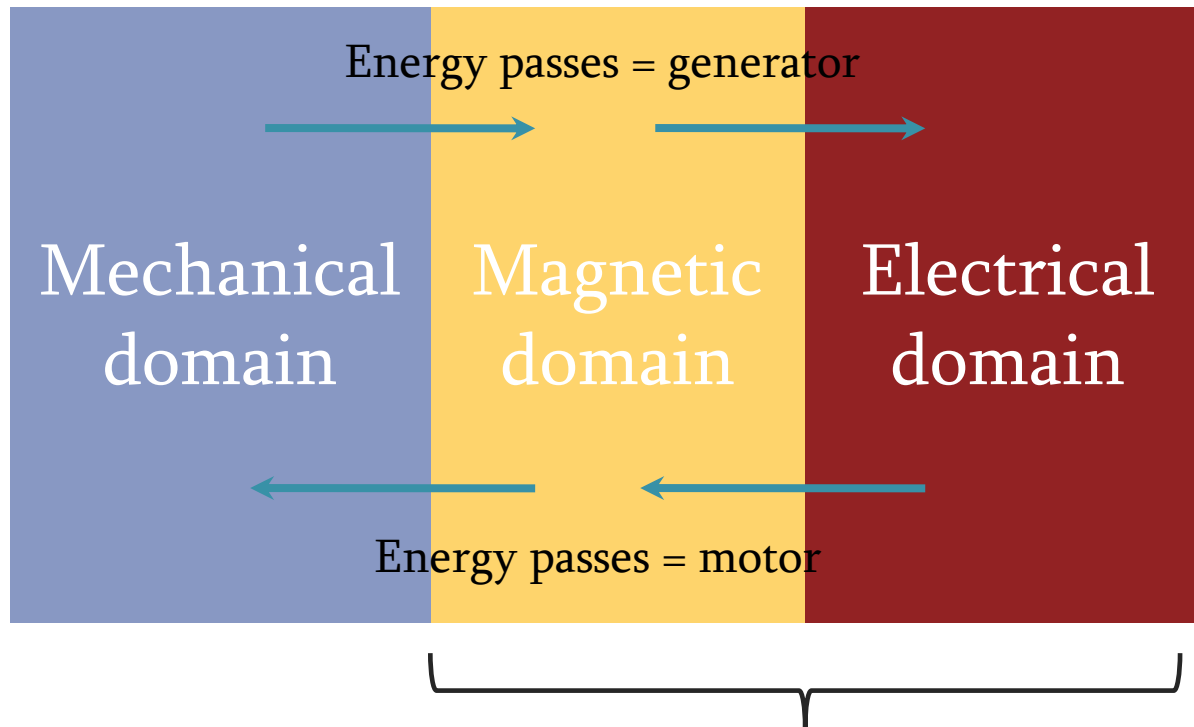
DIRECT CURRENT (DC) MACHINES Fundamentals

- **Generator action:** An emf (voltage) is induced in a conductor if it moves through a magnetic field.
- **Motor action:** A force is induced in a conductor that has a current going through it and placed in a magnetic field
- Any DC machine can act either as a generator or as a motor.

Basic Parts of DC machine



Stator	Stationary part of the machine with poles that carries field winding
Rotor	The rotating part of the machines that carries the armature winding



Principle similar to of transformer;

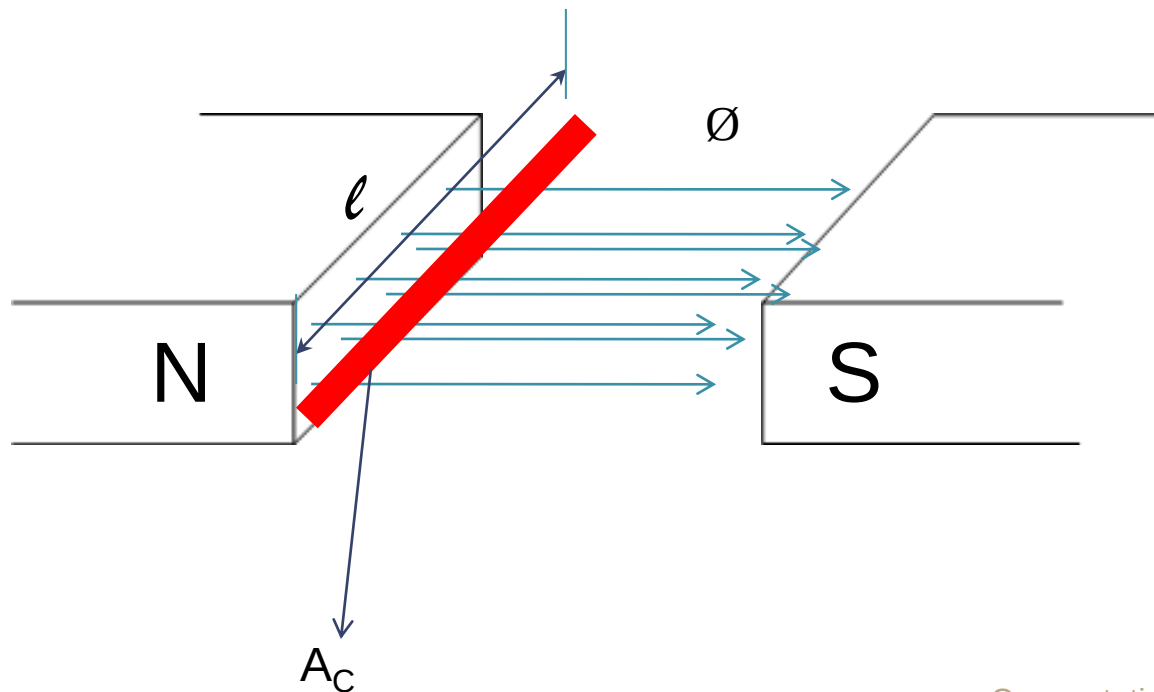
FARADAY'S LAW

Commutation

- Is the process of converting the ac voltages and currents in the rotor of a dc machine to dc voltages and currents at its terminal.
- The basic technique of commutation will be explained for a simple rotating loop.

2. Induced voltage and developed torque

Faraday's Law of Electromagnetism w. r. t conductor



Induced voltage for one conductor

$$e = B \ell v$$

Induced voltage for one conductor

$$e_{ind} = Blv$$

where

B = magnetic flux density (T)

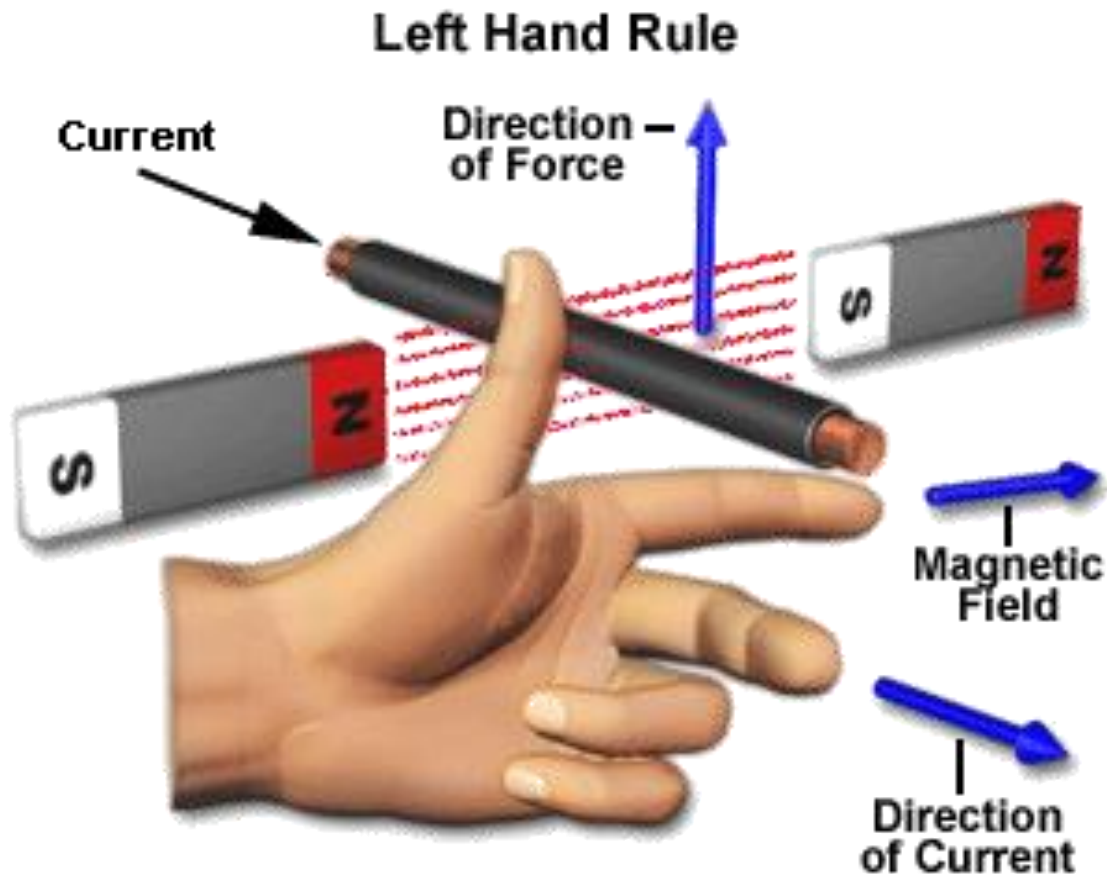
l = length of conductor (m)

v = velocity of the conductor (ms^{-1})

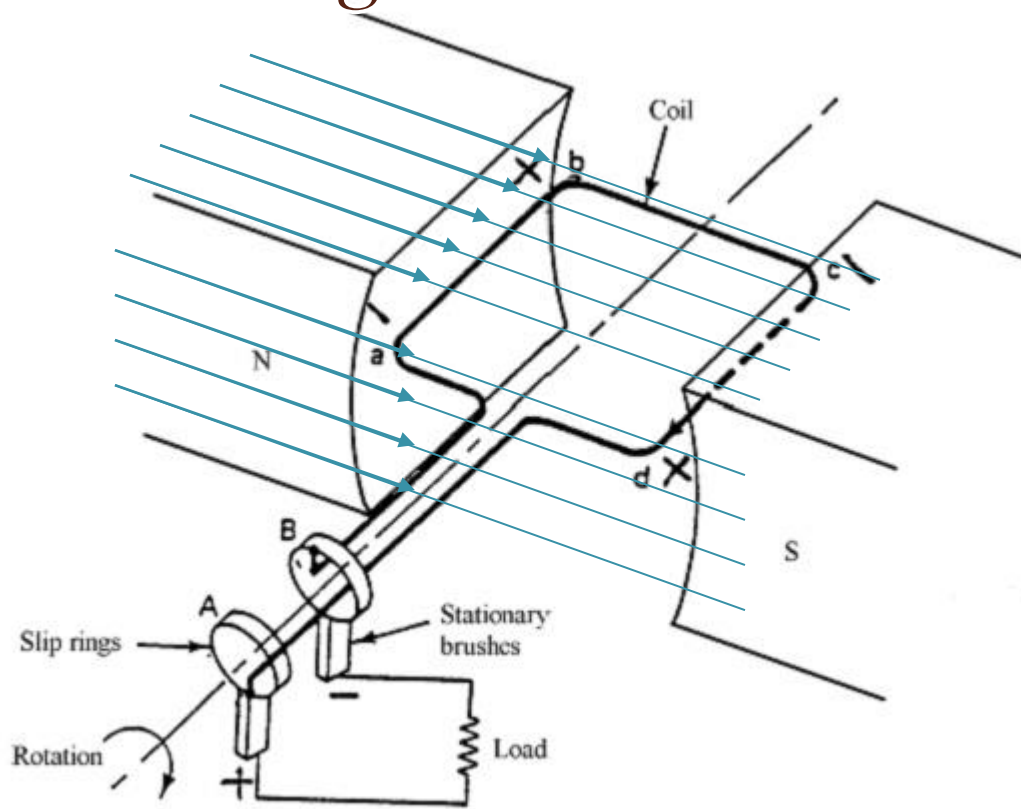
The induced voltage depends on three factor:

1. The flux, Φ in the machine
2. The speed ω_m of the machine's rotor
3. A constant depending on the construction of the machine

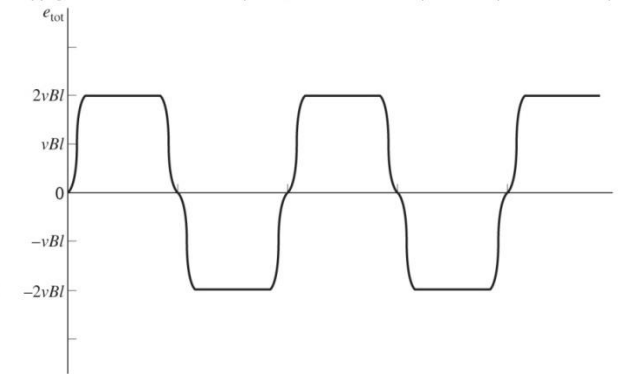
Fleming's Left Hand Rule



Voltage Induced in a Rotating Loop



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The total induced voltage on the loop e_{ind} is given by;

$$e_{ind} = e_{ba} + e_{cb} + e_{dc} + e_{ad}$$

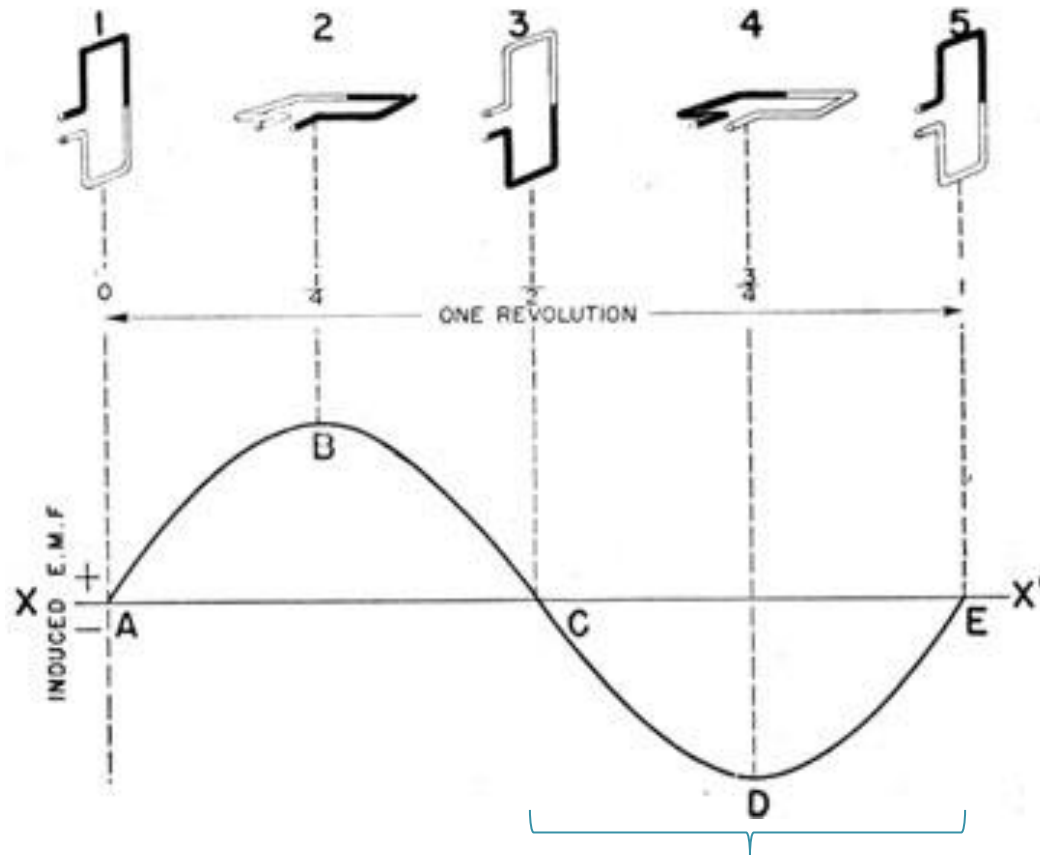
$$e_{ind} = vBl + 0 + vBl + 0$$

$$e_{ind} = 2vBl$$

Induced voltage for a loop

$$e_{ind} = 2B \ell v$$

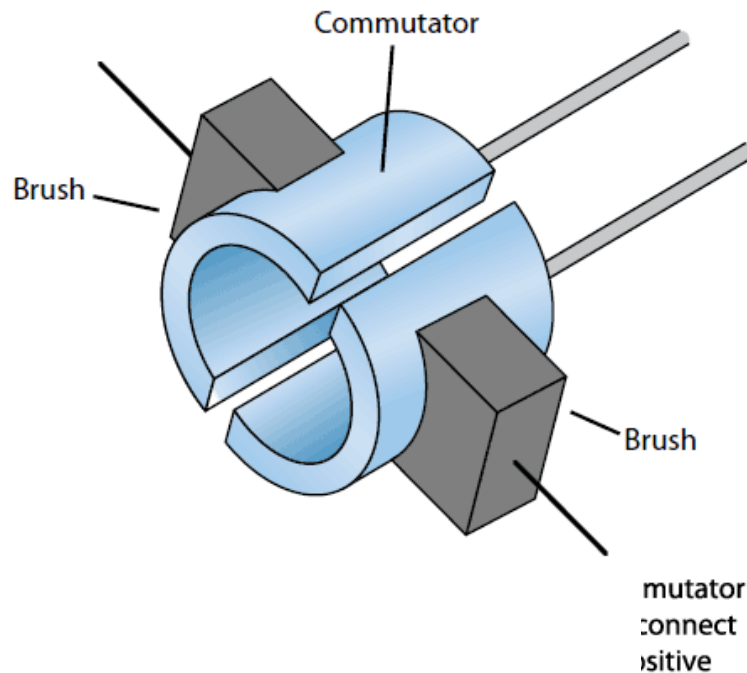
The output voltage from a general AC machine



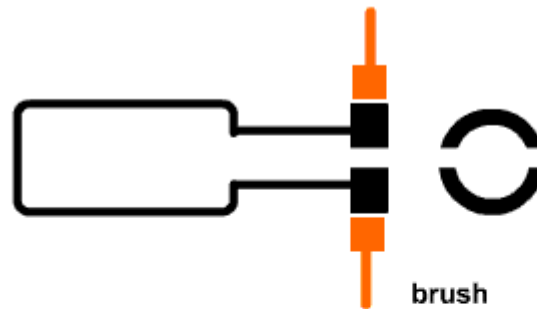
Current obtained from a simple generator reverses its direction after every half revolution. Such current undergoing periodic reversals is known as alternating current (A.C).

How to get DC?

Slip- rings are replaced by split- rings commutator.

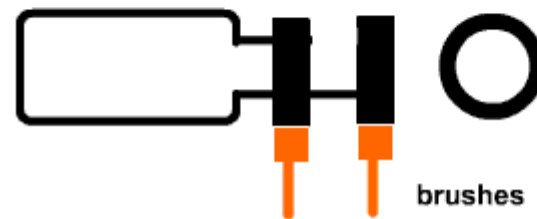


DC

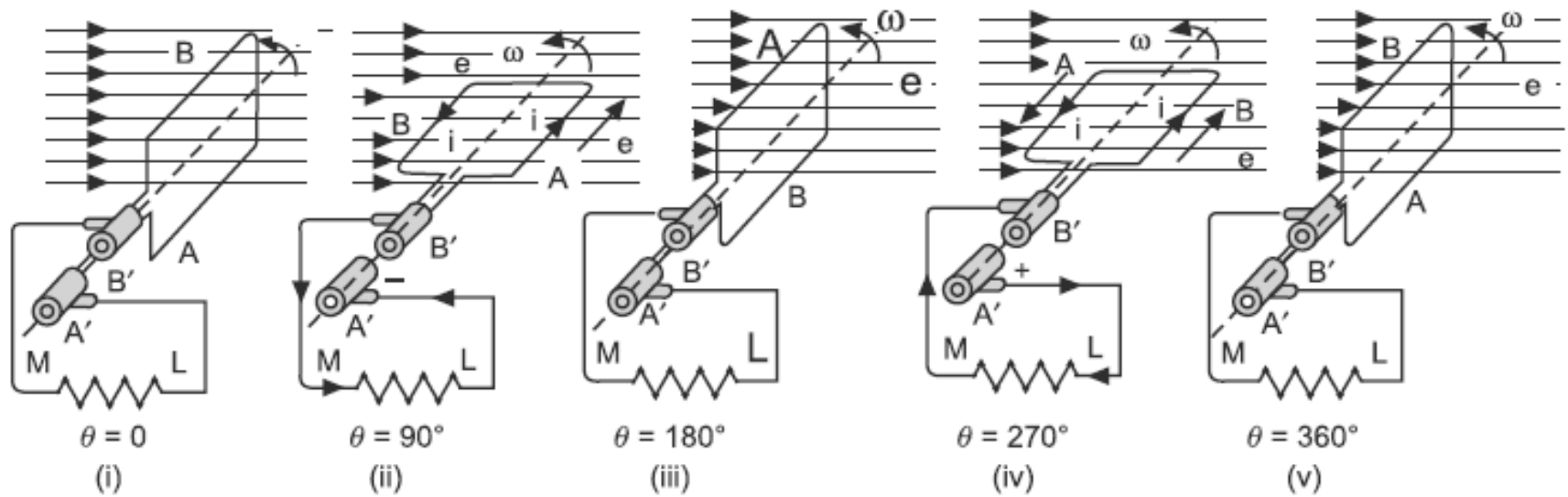


DC with split-ring commutator

AC



AC with two rings



(a) Load connected through slip-rings

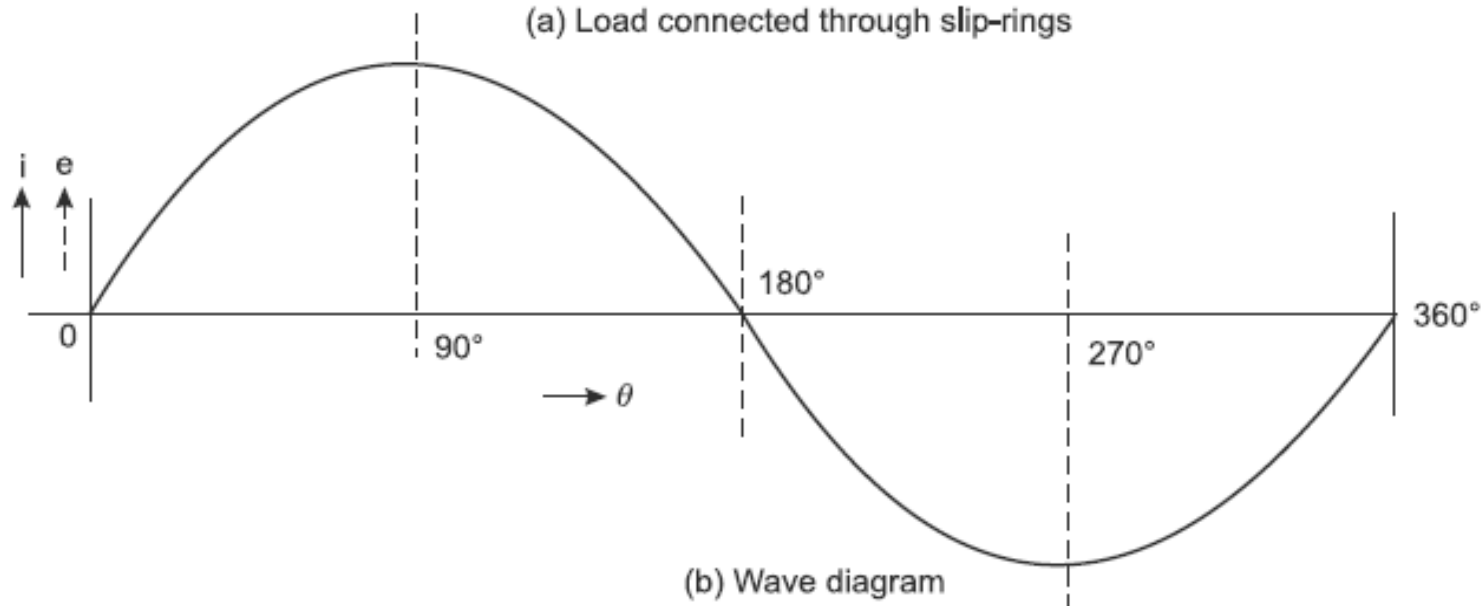
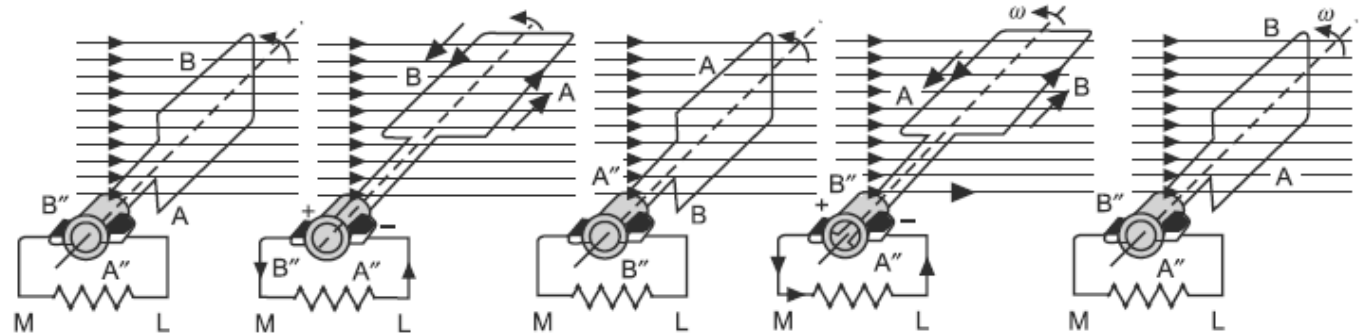
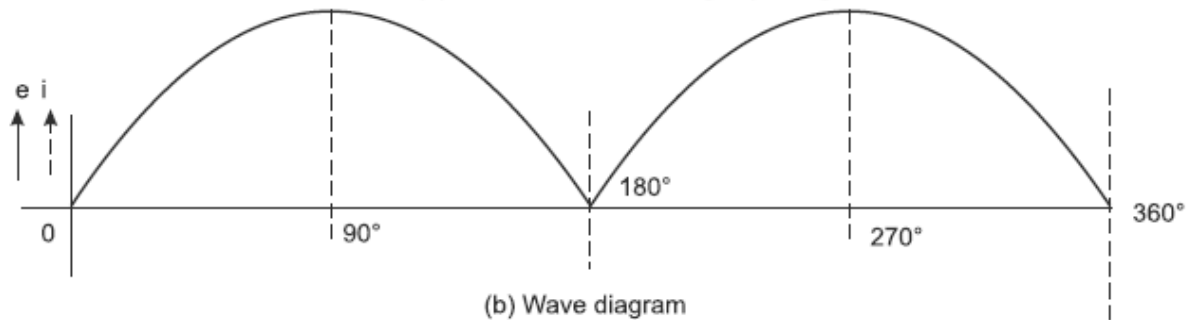


Fig. 4.8 Generated emf for external circuit connected through slip rings

The output voltage from a DC machine



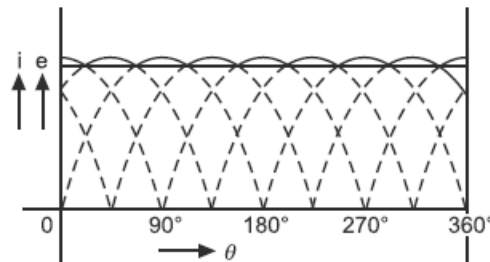
(a) Load connected through split ring



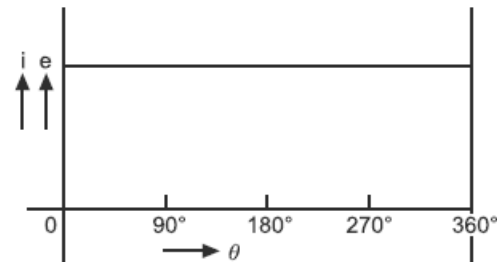
(b) Wave diagram

Fig. 4.9 Generated emf for external circuit connected through split ring

Thus, in actual machine working as a generator, the function of commutator is to convert the alternating current produced in the armature into **direct current** in the external circuit.



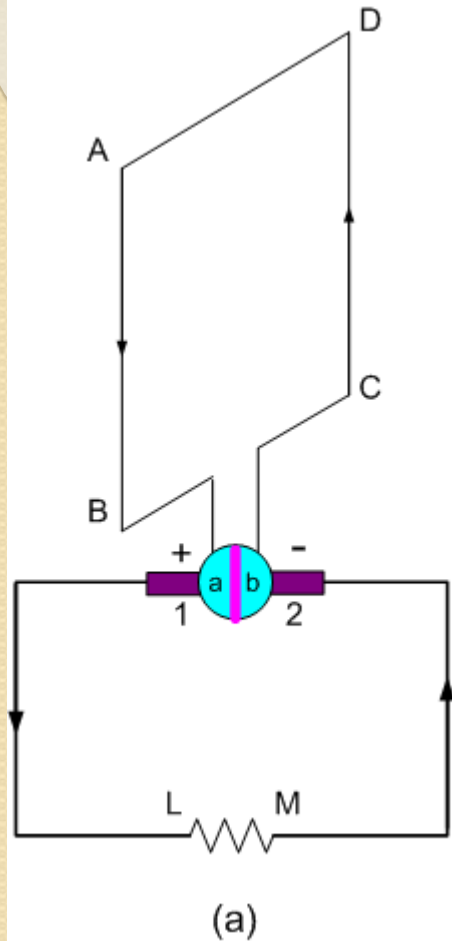
(a) Wave shape of output delivered by number of coils



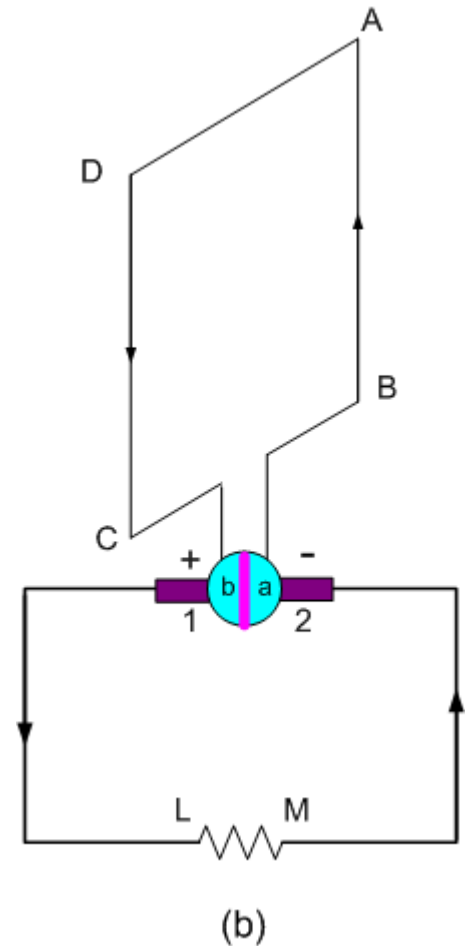
(b) True wave shape of output delivered by a dc generator

Operation

- In the first half revolution, Figure (a), current flows along ABLMCD through brush No. 1 which is in contact with segment 'a' that acts as the positive end of the supply and brush No. 2 and 'b' as the negative end.



- In the next half revolution as in Figure (b), the direction of the induced current in the coil is reversed. But at the same time, the positions of segments 'a' and 'b' are also reversed with the result that brush No. 1 comes in touch with that segment which is positive i.e. segment 'b'.



Induced Torque in Rotating Loop

Segment ab

$$F_{ab} = i (l \times B)$$

$$= ilB$$

$$\tau_{ab} = rF \sin \theta$$

$$= r(ilB) \sin 90^\circ$$

$$= rilB \text{ CCW}$$

Segment bc

$$F_{bc} = i (l \times B)$$

$$= 0$$

$$\tau_{bc} = 0$$

Segment cd

$$F_{cd} = i (l \times B)$$

$$= ilB$$

$$\tau_{cd} = rF \sin \theta$$

$$= r(ilB) \sin 90^\circ$$

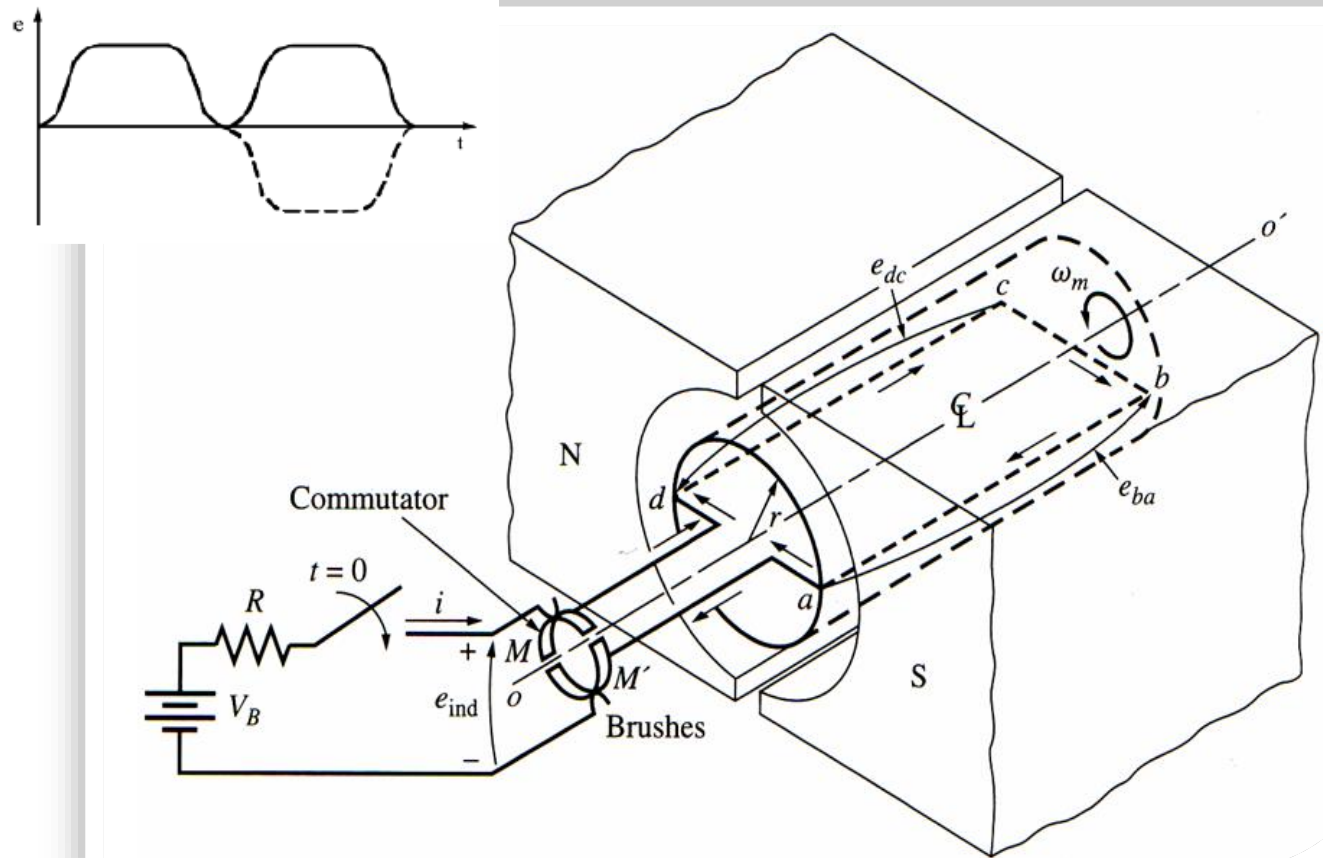
$$= rilB \text{ CCW}$$

Segment da

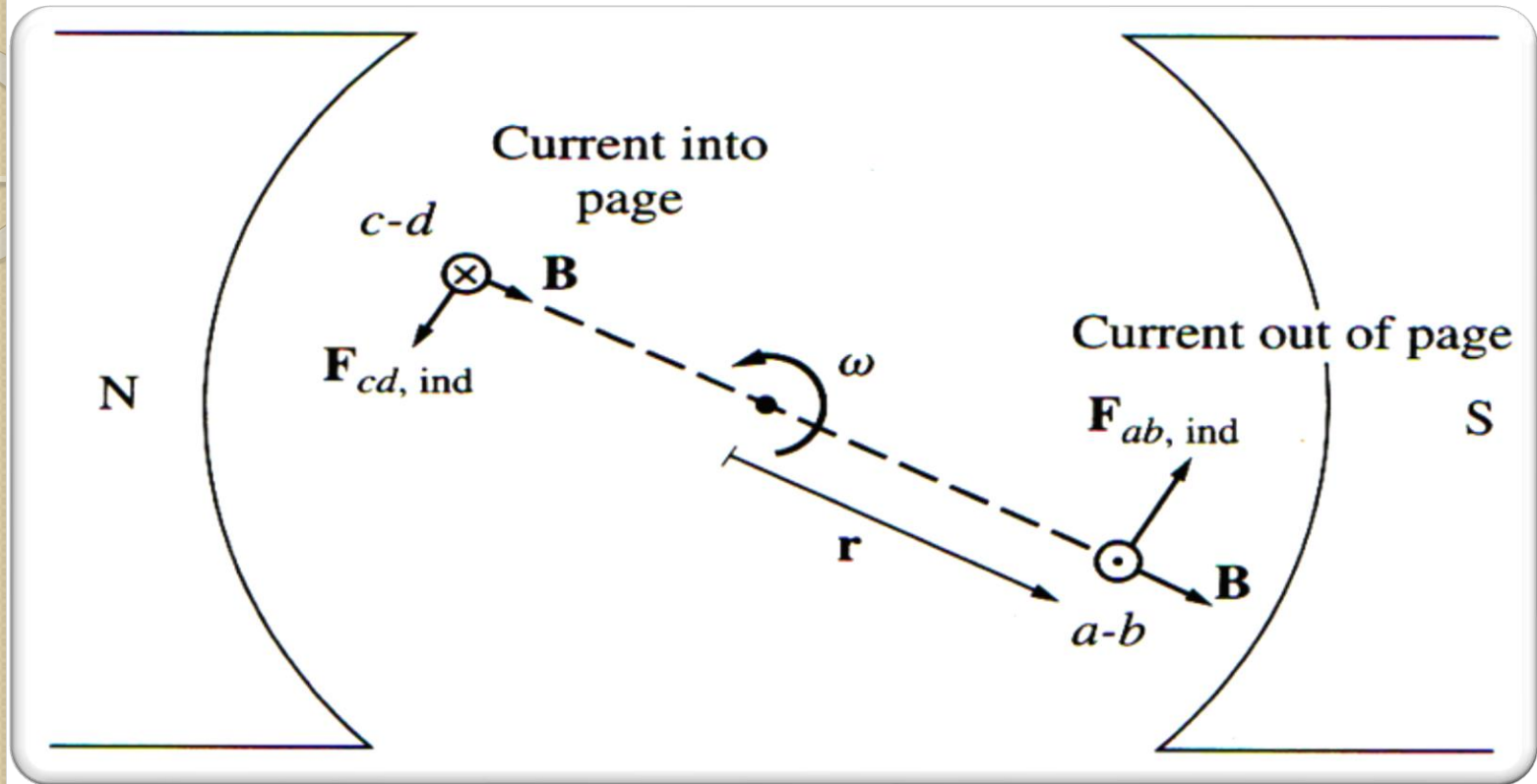
$$F_{da} = i (l \times B)$$

$$= 0$$

$$\tau_{da} = 0$$



Developed torque in a rotating loop



$$\tau_{ind} = 2rilB$$

$$\tau_{ind} = \frac{2}{\pi} \phi i$$

Simplified. since $A_p \approx \pi r l$ and $\phi = A_p B$

Induced torque in the machine

$$\tau_{ind} = 2rilB$$

where

r = radius of the conductor

i = current flow on the conductor

B = magnetic flux density (T)

l = length of conductor (m)

The induced torque depends on three factor:

1. The flux, Φ in the machine
2. The armature (or rotor) current, I_A in the machine
3. A constant depending on the construction of the machine

Related equations

The internal generated voltage in the machine

$$E_A = \frac{ZP}{2\pi a} \Phi \omega_m \quad (\text{in rad/s})$$

$$E_A = \frac{ZP}{60a} \Phi n_m \quad (\text{in rev/min})$$

The current flowing out of machine

$$i = \frac{V_B - e_{ind}}{R} \quad (\text{motor})$$

$$i = \frac{e_{ind} - V_B}{R} \quad (\text{generator})$$

The total induced torque in the machine

$$\tau_{ind} = \frac{ZP}{2\pi a} \Phi I_A \quad (\text{in Nm})$$

The relationship between the induced EMF and torque

$$E_{\text{conductor}} = Blv$$

$$T_{\text{conductor}} = B l i r$$

therefore,

$$\frac{E}{T} = \frac{Blv}{Blir} = \frac{v}{ir}, v = \omega r$$

$$\frac{E}{T} = \frac{\omega}{i}$$

$$\underbrace{Ei}_{\text{electric power}} = \underbrace{T\omega}_{\text{mechanical power}}$$

$$T = \frac{EI}{\omega} = \frac{ZP}{2\pi a} \phi I_A$$

$$T = k_m \phi I_A$$

Where: T is the torque,

3. Power flow and losses in DC machines

DC generators take in mechanical power and produce electric power while DC motors take in electric power and produce mechanical power.

Efficiency;

$$\eta = \frac{P_{out}}{P_{in}} \times 100\% \quad \text{or} \quad \eta = \frac{P_{out} - P_{loss}}{P_{in}} \times 100\%$$

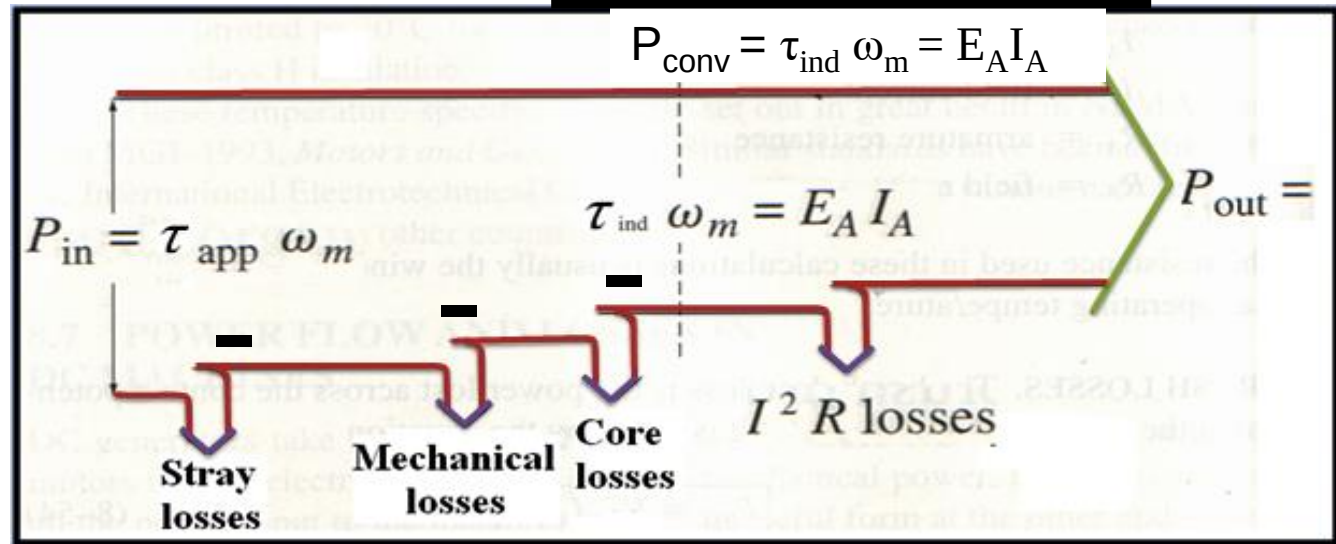
5 categories of losses in DC machine

1. **Copper losses (I^2R)**(All windings have some resistance with current flow in them)
2. **Brush losses** (the contact losses between the brushes made of copper carbon)
3. **Core losses** (hysteresis losses and eddy current losses)
4. **Mechanical losses** (the losses that associated with mechanical effects. Two basic types of mechanical losses: friction & windage. Friction losses caused by the friction of the bearings in the machine. Windage are caused by the friction between the moving parts of the machine and the air inside the motor casing's)
5. **Stray load losses** (cannot placed in one of the previous categories) or Miscellaneous losses

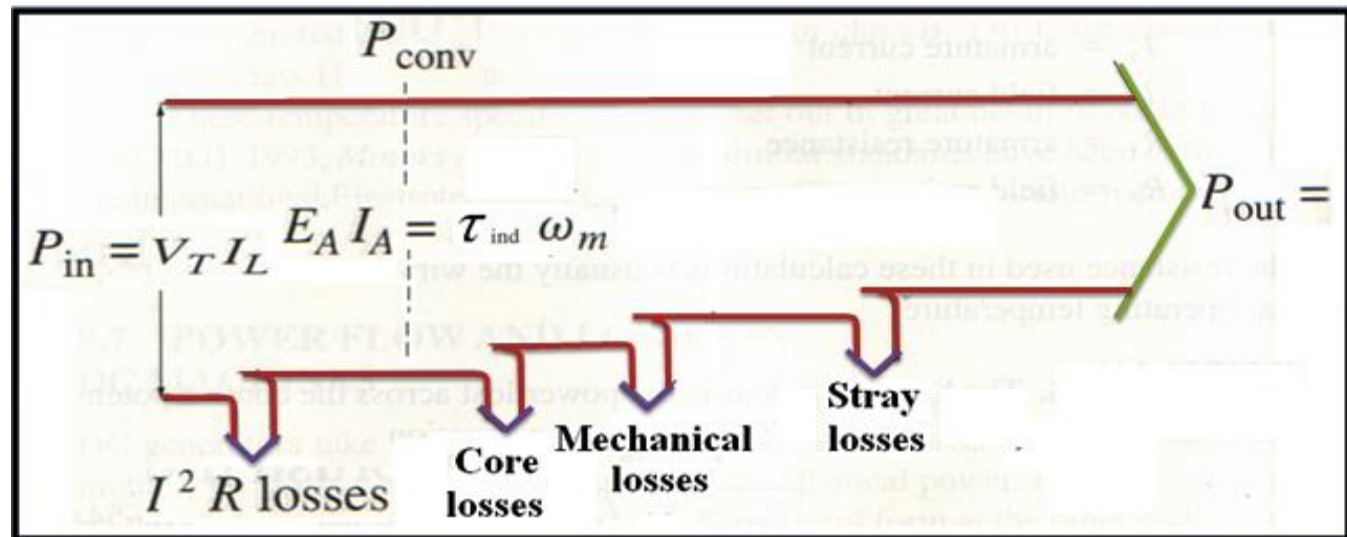
The Power Flow Diagram

$$P_{\text{out}} = E_A I_A - \text{copper loss} - \text{brush loss}$$

For generator



For motor



Equations

Armature loss, $P_A = I_A^2 R_A$

Field loss, $P_F = I_F^2 R_F$

Brush drop loss, $P_{BD} = V_{BD} I_A$



Thank You